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MIL-B-11596B (MU)
8 February 1968
SUPERSEDING
MIL-B-11596A (ORD)
10 November 1960

MILITARY SPECIFICATION

BASE COUPLING, FIRING DEVICE, NON-METALLIC, M2, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for Base Coupling, M2 for use with firing devices.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons; General Specification for
MIL-P-12951	- Primer, Percussion, M39A1, Loading, Assembling and Packing
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Cards
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8826665 - Box, Packing, Ammunition, Base Coupling, Firing Device, Non-Metallic, M2 Assembly
8831456 - Base Coupling, Firing Device, Non-Metallic, M2 Assembly
8845075 - Carton, Packing, Ammunition for Base Coupling, Non-Metallic, M2 Assembly

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8831456 - Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.- Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Base coupling.- The base coupling assemblies shall comply with all requirements specified on Drawing (dwg.) 8831456, and with all requirements specified in applicable specifications.

3.3 Waterproofness.- The base coupling assemblies shall show no evidence of leakage when submerged in water and subjected to an internal air pressure of 10 pounds per square inch gage (p.s.i.g.), minimum, for not less than ten (10) seconds when tested as specified in 4.3.3.1.

3.4 Workmanship.- All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious

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to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.5 First article inspection.- This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define the terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.- Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of 50 assemblies and 50 complete sets of parts (i.e. 50 each of every component and every sub-assembly including at least enough items from each cavity used to produce plastic parts and so identified so that two (2) assemblies can be made up of every possible combination) which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials, (see 4.1.1e). A sample containing known defects will not be submitted unless specifically authorized, by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also

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be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government inspector, shall not be deemed to alter the homogeneity of the lot. Each lot shall contain:

a. Parts of one interfix lot number from one manufacturer.
b. Primers of one interfix lot number from one manufacturer.

Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery shall be homogeneous and of a size convenient to the contractor.

4.3.2 Examinations.- Sampling plans and procedures for the following classification of defects shall be in accordance with MIL-STD-105 except that inspection for critical defects, (and

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majors, when so specified), shall be 100 percent. Contractor's inspection plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 for defects other than critical may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Base (see dwg. 8831457 covering a detail of dwg. 8831456).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Pitch diameter of larger thread.	Gage	01001
102.	Major diameter of larger thread.	Gage	01002
103.	Pitch diameter of smaller thread.....	Gage	01003
104.	Major diameter of smaller thread.....	Gage	01004
105.	Inside diameter of primer cavity, maximum (max.).....	Gage	01005
106.	Diameter of flash hole.....	Gage	01006
107.	Depth of primer cavity.....	Gage	01007
108.	Length to basic diameter.....	Gage	01008
109.	Flash hole obstructed.....	Visual	01009
Minor:	AQL 2.50 percent		
201.	Total length.....	Gage	01010
202.	Length to flange from larger thread face.....	Gage	01011
203.	Diameter of shipping cup end....	Gage	01012
204.	Outside diameter below smaller threads.....	Gage	01013
205.	Ribs missing.....	Visual	01014

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4.3.2.2 Nut (see dwg. 8821458 covering a detail of dwg. 8831456).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Pitch diameter of external thread	Gage	02001
102.	Major diameter of external thread	Gage	02002
103.	Pitch diameter of internal thread, max	Gage	02003
104.	Minor diameter of internal thread, max	Gage	02004
105.	Inside diameter (shipping cup side), max	Gage	02005
Minor:	AQL 1.50 percent		
201.	Depth to basic diameter	Gage	02006
202.	Depth to bottom of thread cavity	Gage	02007
203.	Ribs missing	Visual	02008

4.3.2.3 Assembly (see dwg. 8831456).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Primer missing or loose	Visual-Manual	03001
102.	Primer cocked (incorrectly seated)	Visual	03002
103.	Compound or other foreign matter on primer	Visual	03003
104.	Primer cracked, punctured or with deep scratch	Visual	03004

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105.	Excessive compound.....	Visual	03005
106.	Thread damaged.....	Visual	03006
107.	Crack, split or deep cut.....	Visual	03007
108.	Torque of assembled nut to base exceeds limit.....	Gage	03008

Minor: AQL 1.50 percent

201.	Shipping cup missing or damaged (cracked or punctured).....	Visual	03009
202.	Evidence of poor workmanship (see 3.4).....	Visual	03010
203.	Gasket missing or damaged.....	Visual	03011

4.3.2.4 Carton, sealed (see dwg. 8845075).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed.....	Visual	04001
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface for at least 1/2 inch around entire carton.....	Visual	04002
Minor:	AQL 1.00 percent		
201.	Contents move when carton is shaken.....	Manual	04003
202.	Marking misleading or unidentifiable.....	Visual	04004

4.3.2.5 Box, sealed wooden packing (see dwg. 8826665).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 percent		
101.	Hardware missing, broken or loose	Visual-Manual	05001
102.	Strapping missing, broken or loose	Visual-Manual	05002
103.	Contents exposed	Visual	05003
104.	Board broken or split	Visual	05004
Minor:	AQL 2.50 percent		
201.	Hardware improperly engaged	Visual-Manual	05005
202.	Contents loose (when box is tilted)	Manual	05006
203.	Strapping improperly assembled	Visual	05007
204.	Car seal missing, unsealed or improperly positioned	Visual	05008
205.	Marking misleading or unidentifiable	Visual	05009

4.3.3 Testing

4.3.3.1 Waterproofness (see 3.3) - Major defect - Code No. 06001.- The sampling plan for this test shall be in accordance with MIL-STD-105 using Code Letter K and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. Any base coupling assembly which fails to comply with the specified requirements shall be classed defective. Samples complying with the requirements shall be returned to the lot.

4.3.3.2 Check test for possible deterioration of primers - Major defect - Code No. 07001.- If the total time between original acceptance of any primer lot and the assembly of that lot into the base coupling assemblies exceeds two (2) years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test specified in MIL-P-12951 immediately before the primer lot is assembled into the coupling base assemblies. This test shall be performed by the Government inspector at the facility assembling the primers into the coupling base assemblies and shall be conducted at Government expense without cost to the contractor who loaded the primer or the contractor

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assembling the primer into the base coupling assemblies. Furthermore this test shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the primer lot was under the jurisdiction of either contractor.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 8831456 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.2.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the EL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed on the EL shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.- The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Waterproofness.- The base coupling assembly shall be assembled to an internal air test fixture with a torque not to exceed one inch-pound. It shall then be completely submerged in water and subjected to a minimum internal air pressure of 10 p.s.i.g. for not less than 10 seconds. Observation shall be made for evidence of leakage (air bubbles) to determine compliance with 3.3.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- Base Coupling, Firing Device, Non-Metallic M2 shall be packaged in accordance with dwg. 8845075.

5.1.2 Level C.- Packaging shall be as specified in 5.1.1.

5.2 Packing

5.2.1 Level A.- Base Coupling, Firing Device, Non-Metallic M2 shall be packed in accordance with dwg. 8826665.

5.2.2 Level C.- Packing shall be as specified in 5.2.1.

5.3 Marking.- Marking shall be in accordance with dwgs. 8826665 and 8845075.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

c. Submission of first article quantity.-Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out, may be obtained from the contracting officer.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

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6.3.1 Equipment Lists.-Equipment part or identifying lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 through 6.3.6. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.5 and 6.3.6 will be designated in the EL by the following codes. Other inspection equipment listed on the EL and not coded are defined in 6.3.4.

MU - Army design, mandatory for use.
CD - Contractor design and/or commercial equipment on contractor format. (Requires submission for Government approval as specified in 6.3.7).

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 Envelope drawing.-Envelope drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of envelope drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of envelope drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high

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contrast reproductions. Contractors shall submit three copies of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, for loaded items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey
ATTN: SMUPA-ND5
SMUPA-DC

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A-1079

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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